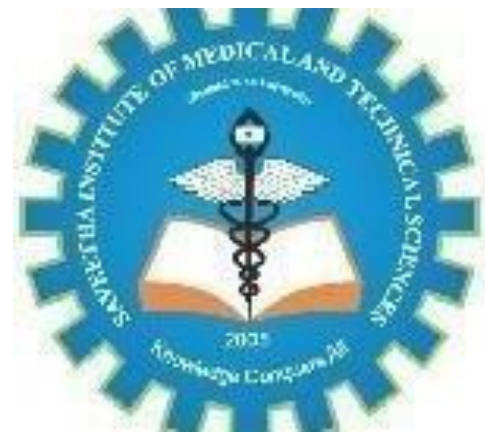




SIMATS SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES



Regular Expressions for Real-World Text Processing

DSA0302-Natural Language Processing

Presented by:

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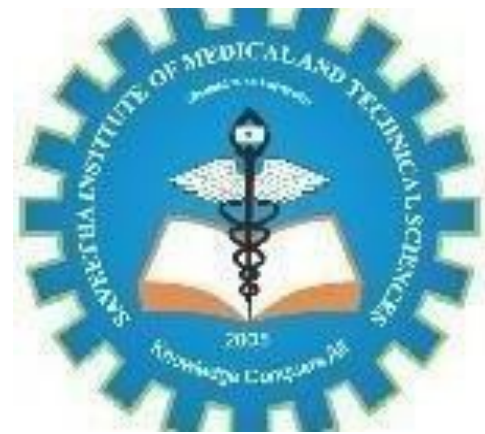
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INTRODUCTION

- Regular Expressions (Regex) are patterns used to search and process text.
- They help computers identify specific text formats quickly.
- Regex is widely used in NLP, web development, cybersecurity, and data analysis.
- It simplifies complex text-processing tasks and improves efficiency.
- Regex helps automate repetitive tasks such as searching, validating, and extracting information from text.



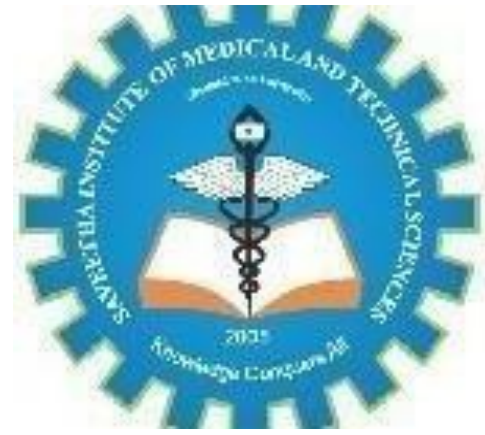
Regular Expressions

- A Regular Expression is a sequence of characters used to define a search pattern.
- It helps in matching, finding, replacing, and extracting text.
- A Regular Expression is a sequence of characters used to define a search pattern.

Symbol	Meaning
.	Matches any character
*	Zero or more occurrences
+	One or more occurrences
?	Optional character
\d	Any digit (0-9)



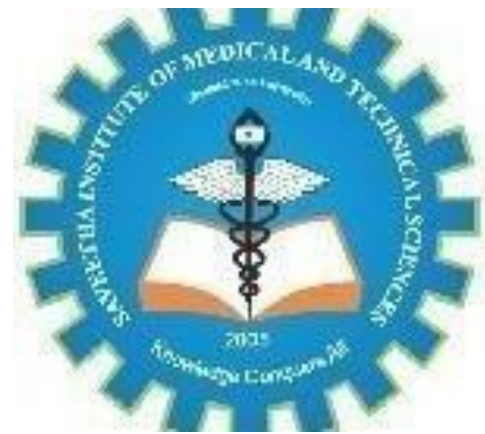
Data Validation Using Regex



- Ensures users enter information in the correct format.
- Prevents invalid data from being stored in databases.
- Validates emails, phone numbers, passwords, and IDs.
- Example: `^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$`
- Commonly used in registration forms and online applications.
- EX: Banking Systems
- University Admission Portals
- Examples:
- Phone Number Validation: `---^\d{10}$`
- Password Validation: `--^(?=.*[A-Z])(?=.*\d).\{8,\}$`



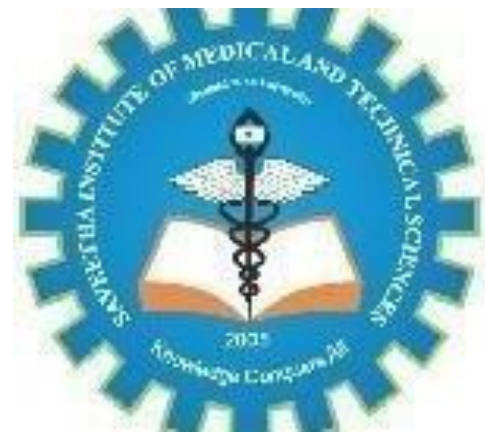
Text Extraction and Information Retrieval



- Regex helps extract useful information from unstructured text.
- Extracts useful information from large text documents.
- Identifies emails, dates, URLs, and phone numbers.
- Reduces manual effort in data collection.
- Examples:
- Extract Email Addresses:
 - `[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}`
- Extract URLs:
 - `https?://\S+`



Log Analysis and NLP Applications



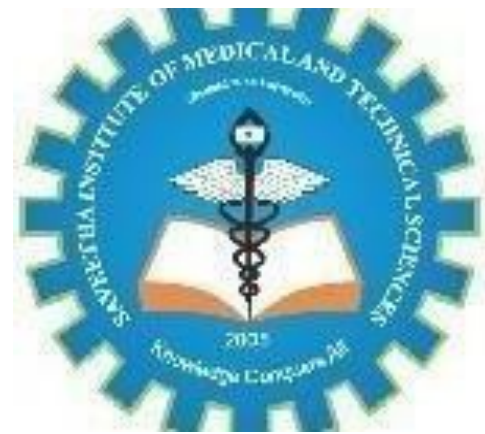
Organizations generate thousands of log records daily.

Example Log

- INFO: User Login Successful
- ERROR: Database Connection Failed
- WARNING: Low Disk Space
- **Regex Pattern**
- ERROR:.*
- **Uses in Log Analysis:**
- Detect System Errors
- Monitor Security Threats
- Identify Performance Issues
- **NLP Applications of Regex:**
- Tokenization (splitting text into words)
- Chatbots and Virtual Assistants
- Named Entity Recognition



IMPLEMENTATION



```
import re
```

```
text = """
```

```
Contact: student@gmail.com
```

```
Phone: 9876543210
```

```
Date: 15-07-2026
```

```
"""
```

```
email = re.findall(r'[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}', text)
```

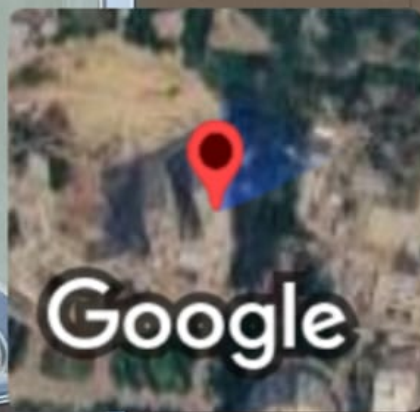
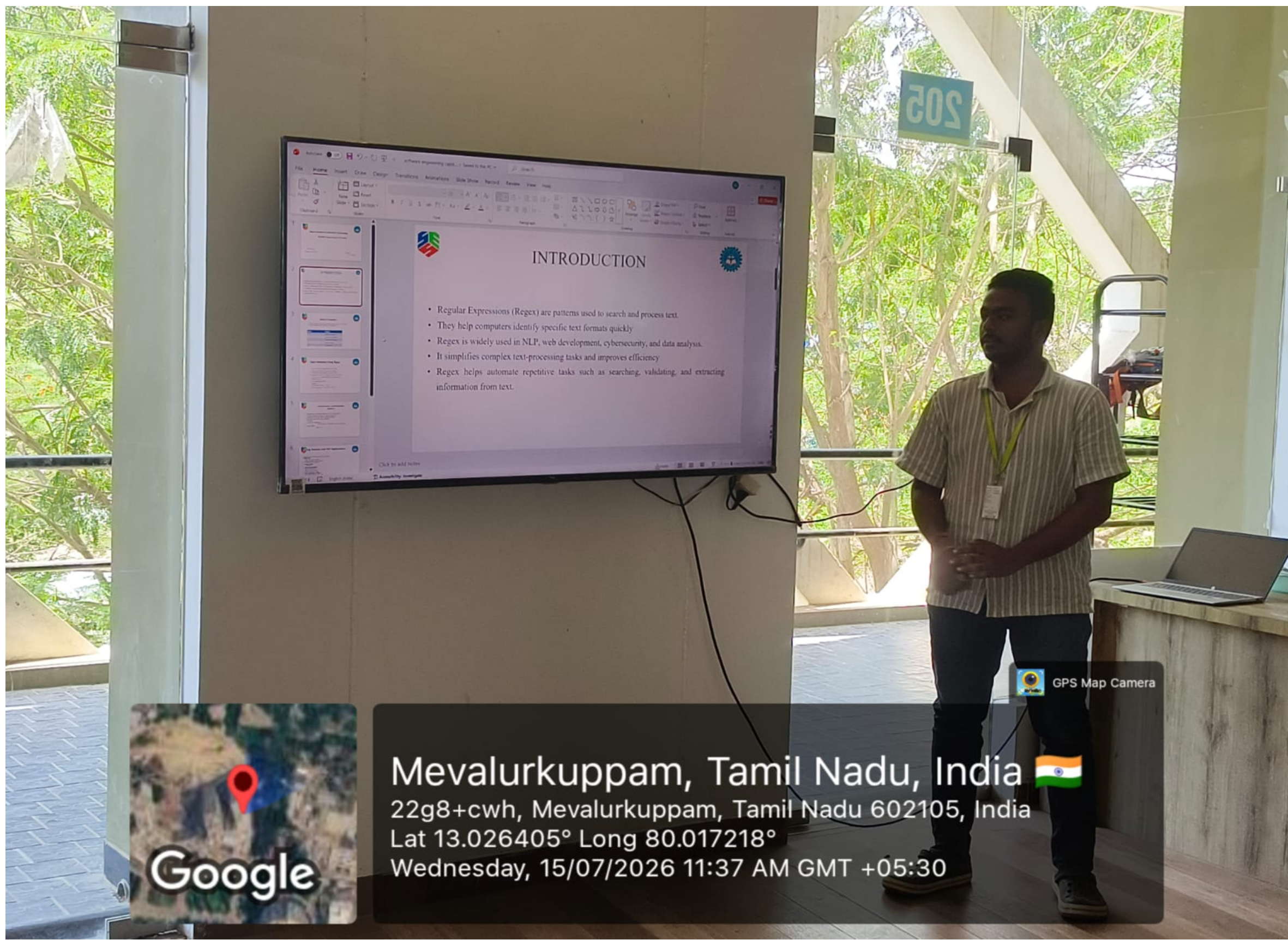
```
phone = re.findall(r'\d{10}', text)
```

```
date = re.findall(r'\d{2}-\d{2}-\d{4}', text)
```

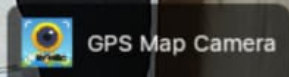
```
print("Email:", email)
```

```
print("Phone:", phone)
```

```
print("Date:", date)
```

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THANK YOU